PRODUCT DESIGN AND PRODUCTION SYSTEMS

Production systems are used for production of goods and services and therefore there's a need to design a system of producing required goods and services.

PRODUCT DESIGN:

The process of creating or improving a product by learning what consumers want and examining similar products that are already available. **OR** Product design is the process of efficient and effective idea generation and development with the goal of creating new products. Product design includes all the engineering and industrial design work that goes into developing a product from the initial concept to production, which ensures that it works reliably, is cost effective to manufacture and looks good.

STRATEGIES FOR NEW PRODUCT INTRODUCTION.

There are ***3 fundamental different ways*** to introduce new products. These approaches include:

1. **Markets pull**. The market is a primary basis for determining the products a firm should make, with little regard to existing technology.
   Key aspects of market pull include:
   - **Customer-Led Development:** Products are created in response to identified consumer needs or societal pressures.
   - **High Demand:** It often involves products that generate significant "buzz".
   - **Examples:** The development of smartphones with selfie cameras or the creation of eco-friendly packaging.
   - **Market Validation:** It helps confirm a product-market fit, reducing the risk of failure.
2. **Technology push**. Technology is the primary determinant of the products that the firm should make with little regard for the market. The firm should pursue a technology-based advantage by developing superior products. It's where new products are developed based on scientific research and technical advancements rather than identified market demand.
   Key Characteristics & Aspects
   - **Driver:** Primarily driven by R&D, new technologies, or manufacturing capabilities.
   - **Process:** A linear, top-down approach where technology is developed first, then pushed into the market.
   - **Market Need:** There is no initial, clearly defined consumer demand or problem to be solved.
   - **Challenges:** It requires significant effort to create a market, convince consumers of the product's value, and overcome potential, initial lack of interest.
   - **Goal:** Aims for radical innovation and the creation of entirely new markets. E.g smartphones, ipads/tabs etc touch screen laptops
3. **Interfunctional view**. The view holds that products should not only fit the market needs but have a technical advantage as well. To accomplish this, all functions (e.g., marketing, engineering, operations and finance), should cooperate to design the new products needed by the firm.

Technology Push vs. Market Pull

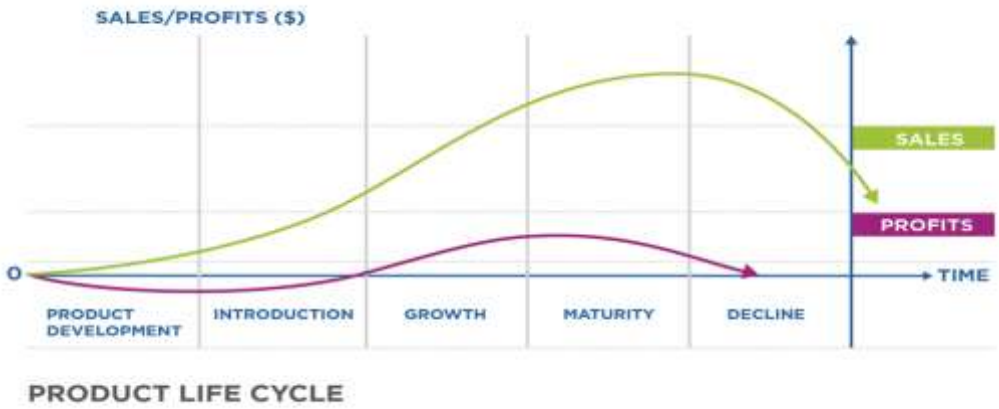
Feature	Technology Push	Market Pull (Demand Pull)
Trigger	Scientific discovery or new R&D	Identified customer needs or problems
Strategy	"If we build it, they will come"	"The customer needs this, let's build it"
Risk	High market uncertainty	Lower market risk; higher competition
Result	Radical/Transformative innovation	Incremental/Sustaining innovation

THE PRODUCT LIFE CYCLE:

As consumers, we buy millions of products every year. And just like us, these products have a life cycle. Older, long-established products eventually become less popular, while in contrast, the demand for new, more modern goods usually increases quite rapidly after they are launched.

Because most companies understand the different product life cycle stages, and that the products they sell all have a limited lifespan, the majority of them will invest heavily in new product development in order to make sure that their businesses continue to grow.

Product Life Cycle Stages



The product life cycle has 4 very clearly defined stages, each with its own characteristics that mean different things for business that are trying to manage the life cycle of their particular products.

Introduction Stage: This stage of the cycle could be the most expensive for a company launching a new product. The size of the market for the product is small, which means sales are low, although they will be increasing. On the other hand, the cost of things like research and development, consumer testing, and the marketing needed to launch the product can be very high, especially if it's a competitive sector.

Growth Stage: The growth stage is typically characterized by a strong growth in sales and profits, and because the company can start to benefit from economies of scale in production, the profit margins, as well as the overall amount of profit, will increase. This makes it possible

for businesses to invest more money in the promotional activity to maximize the potential of this growth stage.

Maturity Stage: During the maturity stage, the product is established and the aim for the manufacturer is now to maintain the market share they have built up. This is probably the most competitive time for most products and businesses need to invest wisely in any marketing they undertake. They also need to consider any product modifications or improvements to the production process which might give them a competitive advantage.

Decline Stage: Eventually, the market for a product will start to shrink, and this is what's known as the decline stage. This shrinkage could be due to the market becoming saturated (i.e. all the customers who will buy the product have already purchased it), or because the consumers are switching to a different type of product. While this decline may be inevitable, it may still be possible for companies to make some profit by switching to less-expensive production methods and cheaper markets.

Product Life Cycle Examples

It's possible to provide examples of various products to illustrate the different stages of the product life cycle more clearly. Here is the example of watching recorded television and the various stages of each method:

- Introduction – *electric cars, new gaming consoles, wearable fitness trackers etc*
- Growth – *Smart phones, smart watches, popular streaming services*
- Maturity – *Coca cola, pepsi, personal computers, washing machines, LCD TVs*
- Decline – *fax machines, landline phones, VHS tapes, old style digital cameras*

The idea of the product life cycle has been around for some time, and it is an important principle manufacturers need to understand in order to make a profit and stay in business.

However, the key to successful manufacturing does not just understand this life cycle, but also proactively managing products throughout their lifetime, applying the appropriate resources and sales and marketing strategies, depending on what stage products are at in the cycle.

COMMON CHARACTERISTICS OF EACH STAGE

0. Product development stage	**1.** Investment is made **2.** Sales have not begun **3.** New product ideas are generated, operationalized, and tested
1. Market introduction stage	**1.** Costs are very high **2.** Slow sales volumes to start **3.** Little or no competition **4.** Demand has to be created **5.** Customers have to be prompted to try the product **6.** Makes little money at this stage
2. Growth stage	**1.** Costs reduced due to economies of scale **2.** Sales volume increases significantly **3.** Profitability begins to rise

	4. Public awareness increases **5.** Competition begins to increase **6.** Increased competition leads to price decreases
3. Maturity stage	**1.** Costs are lowered as a result of increasing production volumes **2.** Sales volume peaks and market saturation is reached **3.** New competitors enter the market **4.** Prices tend to drop due to the proliferation of competing products **5.** Brand differentiation and feature diversification is emphasized to maintain or increase market share **6.** Profits decline
4. Decline stage	**1.** Costs increase due to some loss of economies of scale **2.** Sales volume declines **3.** Prices and profitability diminish **4.** Profit becomes more a challenge of production/distribution efficiency than increased sales

DESIGNING SERVICES AND SERVICE PROCESS TECHNOLOGY

SERVICE DESIGN

Definition: Service design is the activity of planning and organizing a business's resources (people, products, and processes) in order to **(1)** directly improve the employee's experience, and **(2)** indirectly, the customer's experience.

FIVE PRINCIPLES OF SERVICE DESIGN

1. **User-Centered:** People are at the center of the service design.
2. **Co-Creative:** Service design should involve other people, especially those who are part of a system or a service.
3. **Sequencing:** Services should be visualized by sequences, or key moments in a customer's journey.
4. **Evidencing:** Customers need to be aware of elements of a service. Evidencing creates loyalty and helps customers understand the entire service experience.
5. **Holistic:** A holistic design takes into account the entire experience of a service. Context matters

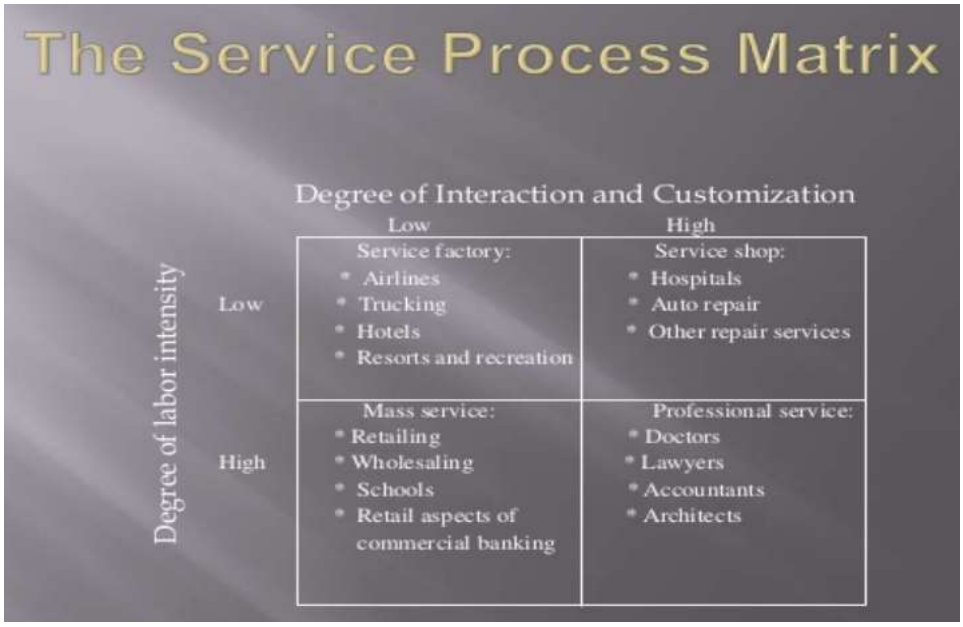
CHARACTERISTICS/FEATURES OF SERVICES

- **Services are intangible.** Designing a service requires one to describe what the customer is supposed to experience, which may be difficult. Designers begin by compiling psychographics. Understanding the customer and his or her expectations is essential in designing good service.

- **Service output is variable**. I.e. it is not fixed because there are many service providers serving a variety of customers who have varying needs. Despite the varying needs, the service experience should be consistent. Companies, which have tried to maintain consistency in service provision, include McDonald's, Stanbic bank, standard chartered bank etc. They maintain this consistency by carrying out extensive employee training, setting operating procedures, and standardized materials, equipment, and physical environments.
- **Services have higher customer contact:** - Ensuring that the service encounter is positive is part of service design. There is no opportunity for testing and reworking.
- **Services are perishable:** - Because they cannot be stored, the timing and location of delivery are important. Service design should design what is to be delivered, when and where.
- **It is difficult for the consumers to separate the service from its delivery;** hence service design and process design must occur concurrently. Service design should also determine how the service should be provided.
- **They can easily be copied by competitors** and as such new service introductions and service improvements occur more rapidly than product improvements.
- **Largely labour intensive**
- **Difficult to measure** output
- **Production and consumption occur** concurrently.

SERVICE CHARACTERISTICS MATRIX

A 2-by-2 matrix is used to portray some characteristics and requirements of different service delivery systems. The matrix divides services into four groups, with the complexity of the service as one dimension and the degree of service customization as the other as below:



`Detail

- The complexity of the service represents the degree of knowledge and skill or capital investment that the service provider must possess in relation to the amount that the typical consumer might have.
- Services on the left of the matrix require a great deal more training and/or capital investment to perform because the services are more complex to perform and require knowledge or equipment that cannot be readily obtained by the typical consumer.
- Services in the upper half of the matrix are more customized or personalized in that they are designed specifically for the individual needs of a particular consumer.
- The services in the lower half of the matrix are more standardized in that the same service will be received by many consumers and there is a possibility of delivery to large numbers simultaneously.
- Since workers in many services are likely to have contact with customers businesses in all the four quadrants of the matrix select persons who are warm and courteous and have good interpersonal skills for the job that entail significant contact with customers.
- Many of the service businesses in quadrants II and IV are more likely to develop procedures and train employees for reliable, consistent service encounters.
- Workers in quadrant II need to have broader skills and to be more flexible in response to the directions of the consumer.
- The work involved in many of the services in quadrant I require extensive professional training that must be obtained outside the business. Problem-solving and advisory activities often are an important part of these services, so employees must have good perception of conditions and good diagnostic abilities.
- Investment in facilities and equipment is often significant for services in quadrant III. The amount of special training for employees in these businesses may also be high, but economies of scale help reduce the cost to each consumer. These businesses must interpret the needs of the people and offer the appropriate mix of services if they want to attract large numbers of customers.

THE SERVICE PROCESS MATRIX:

A professional service, such as a doctor or lawyer, is highly customized and very labour intensive. A service shop, such as schools and hospitals, is less customized and labour intensive but attentive to individual customers. A mass service, such as retailing and banking, offers the same services to all customers and allows less interaction with the service provider. Services like airlines have a minimal degree of customization and labour intensity they are therefore more like manufactured products and hence best processed by a service factory.

Service factory: Electricity is a commodity available continuously to customers.

Mass service: A retail store provides a standard array of products from which customers choose.

Service shop: A professor interacts with a classroom of students. Although a lecture may be prepared in advance, the students in each class affect its delivery.

Professional Service: A doctor provides personal service to each patient based on extensive training in medicine.

		Degree of Customization	
		Low	**High**
Degree of Labour intensity	**Low**	**Service factory** • Airlines • Hotels • Resorts and recreation	**A service shop** • Hospitals • Repair services
	High	**Mass service** • Retailing • Wholesaling • Schools • Retail aspect of commercial banking	**Professional service** • Doctors • Lawyers • Accountants • Architects

Source: Roger G Schroeder (2007) Operations Management

FEATURES OF A WELL-DESIGNED SERVICE SYSTEM

- Consistent with the firm's strategic focus
- User friendly
- Robust
- Easy to sustain
- Effectively linked between front and back office
- Cost effective
- Visible to customer

TYPES OF PRODUCTION SYSTEMS

Process focused:

Under this type, the products, people and all other resources are organized according to the process of production. The individual product requirements dictate the flow of the items being processed. The process-focused system is suitable for customized products. E.g job-shop

Product focused

Under this, the products, people and other resources are organized according to the products to be produced. This system is suitable for standardized products. E.g mass & continouse prod

TYPES OF PRODUCTION PROCESSES & TECHNOLOGY

Production systems can be classified as Job-shop, Batch, Mass and Continuous production systems.

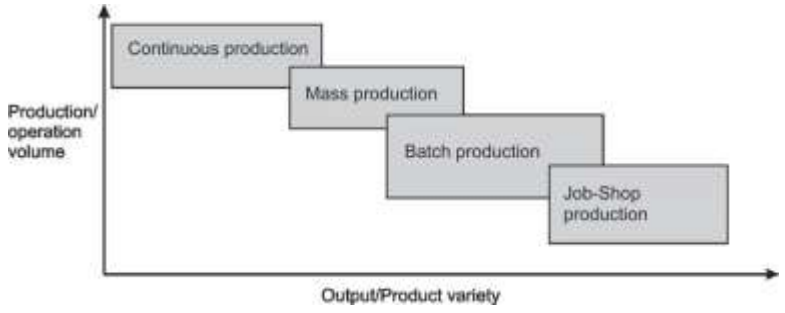


Fig. 1 Classifications of production systems

JOB-SHOP PRODUCTION

Job-shop production are characterized by manufacturing one or few quantity of products designed and produced as per the specification of customers within prefixed time and cost. The distinguishing feature of this is low volume and high variety of products.

A job-shop comprises of general-purpose machines arranged into different departments. Each job demands unique technological requirements, demands processing on machines in a certain sequence. E.g. furniture, fabricators etc

Job-shop Production is characterized by:

- High variety of products and low volume.
- Use of general purpose machines and facilities.
- Highly skilled operators who can take up each job as a challenge because of uniqueness.
- Large inventory of materials, tools, parts.
- Detailed planning is essential for sequencing the requirements of each product, capacities for each work centre and order priorities.

Advantages

Following are the advantages of Job-shop Production:

- Because of general purpose machines and facilities variety of products can be produced.
- Operators will become more skilled and competent, as each job gives them learning opportunities.
- Full potential of operators can be utilized.
- Opportunity exists for Creative methods and innovative ideas.

Limitations

Following are the limitations of Job-shop Production:

- Higher cost due to frequent set up changes.
- Higher level of inventory at all levels and hence higher inventory cost.
- Production planning is complicated.
- Larger space requirements.

BATCH PRODUCTION

American Production and Inventory Control Society (APICS) define Batch Production as a form of manufacturing in which the job pass through the functional departments in lots or batches and each lot may have a different routing. It is characterized by the manufacture of limited number of products produced at regular intervals and stocked awaiting sales. E.g. (bakeries (bread, pastries), clothing manufacturers, pharmaceuticals, textiles etc)

Batch Production is characterized by:

- Shorter production runs.
- Plant and machinery are flexible.
- Plant and machinery set up is used for the production of item in a batch and change of set up is required for processing the next batch.
- Manufacturing lead-time and cost are lower as compared to job order production.

Advantages

Following are the advantages of Batch Production:

- Better utilization of plant and machinery.
- Promotes functional specialization.
- Cost per unit is lower as compared to job order production.
- Lower investment in plant and machinery.
- Flexibility to accommodate and process number of products.
- Job satisfaction exists for operators.

Limitations

Following are the limitations of Batch Production:

- Material handling is complex because of irregular and longer flows
- Production planning and control is complex.
- Work in process inventory is higher compared to continuous production.
- Higher set up costs due to frequent changes in set up.

MASS PRODUCTION

Manufacture of discrete parts or assemblies using a continuous process are called *Mass Production*. This production system is justified by very large volume of production. The machines are arranged in a line or product layout. Product and process standardization exists and all outputs follow the same path e.g. (Automotive industry, soft drinks, textiles, consumer electronics). Mass Production is characterized by

- Standardization of product and process sequence.
- Dedicated special purpose machines having higher production capacities and output rates.
- Large volume of products.
- Shorter cycle time of production.
- Lower in process inventory.
- Perfectly balanced production lines.
- Flow of materials, components and parts is continuous and without any back tracking.
- Production planning and control is easy.
- Material handling can be completely automatic.

Advantages

Following are the advantages of Mass Production:

- Higher rate of production with reduced cycle time.
- Higher capacity utilization due to line balancing.

- Less skilled operators are required.
- Low process inventory.
- Manufacturing cost per unit is low.

Limitations

Following are the limitations of Mass Production:

- Breakdown of one machine will stop an entire production line.
- Line layout needs major change with the changes in the product design.
- High investment in production facilities.
- The cycle time is determined by the slowest operation.

CONTINUOUS PRODUCTION

Production facilities are arranged as per the sequence of production operations from the first operations to the finished product. The items are made to flow through the sequence of operations through material handling devices such as conveyors, transfer devices, etc.

In **Continuous Production**, the process runs 24/7 with virtually no interruptions. The "product" is typically a flow of material rather than individual pieces until the very end of the line.

- **Oil & Gas Refineries:** Crude oil is processed in a constant flow through distillation towers to create gasoline, diesel, and kerosene (a classic Process Production model).
- **Steel Manufacturing:** Blast furnaces must stay hot and running constantly to produce rolls of steel; shutting down is extremely costly and time-consuming.
- **Electricity Generation:** Power plants (nuclear, coal, or hydro) produce a constant "flow" of energy to meet grid demand.
- **Paper Mills:** Large machines process wood pulp into massive continuous rolls of paper at high speeds.
- **Chemical Plants:** Production of fertilizers, plastics, or industrial gases where the chemical reaction must be maintained under specific pressure and temperature.

Continuous Production is characterized by

- Dedicated plant and equipment with zero flexibility.
- Material handling is fully automated.
- Process follows a predetermined sequence of operations.
- Component materials cannot be readily identified with final product.
- Planning and scheduling is a routine action.

Advantages

Following are the advantages of Continuous Production:

- Standardization of product and process sequence.
- Higher rate of production with reduced cycle time.
- Higher capacity utilization due to line balancing.
- Manpower is not required for material handling as it is completely automatic.
- Person with limited skills can be used on the production line.
- Unit cost is lower due to high volume of production.

Limitations

Following are the limitations of Continuous Production:

- Flexibility to accommodate and process number of products does not exist.
- Very high investment for setting flow lines.
- Product differentiation is limited.

DESIGNING THE PRODUCTION SYSTEM

This is the preliminary stage of the production process. It is the art of planning, or creating systems for production and provision of products and services. In designing systems for both services and products, various factors have to be considered and these include:

a) Capital requirements
b) Required skills for the design of the programme
c) Product design
d) Demand for the product or service
e) Safety requirements etc.

Classification of products in the production design system

The process or production design system involves the collecting of human resource, machinery and other factors of production and organizing them in a process to produce the needed services or products.

In designing such a process, products can be classified as follows:

i) Customized products

These are specifically designed to the specifications and needs of the clients. Emphasis is placed in uniqueness, quality, and dependability especially on time of delivery and flexibility of the process to accommodate the different clients.

ii) Standardized products

Are uniform products and are readily available in the inventory. Their prices are also pre-determined.

NEW PRODUCT DEVELOPMENT STAGES

Before a product can embark on its journey through the four product life cycle stages, it has to be developed. New product development is typically a huge part of any manufacturing process. Most organizations realize that all products have a limited lifespan, and so new products need to be developed to replace them and keep the company in business. Just as the product life cycle has various stages, new product development is also broken down into a number of specific phases.

New Product Development

Developing a new product involves a number of stages which typically center on the following key areas:

The Idea: Every product has to start with an idea. In some cases, this might be fairly simple, basing the new product on something similar that already exists. In other cases, it may be something revolutionary and unique, which may mean the idea generation part of the process is much more involved. In fact, many of the leading manufacturers will have whole departments that focus solely on the task of coming up with 'the next big thing'.
Research: An organization may have plenty of ideas for a new product, but once it has selected the best of them, the next step is to start researching the market. This enables them to see if there's likely to be a demand for this type of product, and also what specific features need to be developed in order to best meet the needs of this potential market.
Development: The next stage is the development of the product. Prototypes may be modified through various design and manufacturing stages in order to come up with a finished product that consumers will want to buy.
Testing: Before most products are launched and the manufacturer spends a large amount of money on production and promotion, most companies will test their new product with a small group of actual consumers. This helps to make sure that they have a viable product that will be profitable, and that there are no changes that need to be made before it's launched.
Analysis: Looking at the feedback from consumer testing enables the manufacturer to make any necessary changes to the product, and also decide how they are going to launch it to the market. With information from real consumers, they will be able to make a number of strategic decisions that will be crucial to the product's success, including what price to sell at and how the product will be marketed.
Introduction: Finally, when a product has made it all the way through the new product development stage, the only thing left to do is introduce it to the market. Once this is done, good product life cycle management will ensure the manufacturer makes the most of all their effort and investment.

PROCESS SELECTION WITH BREAK-EVEN ANALYSIS: A TOOL FOR PRODUCT SREENING

Break-even analysis is a technique widely used by production management and management accountants. It is based on categorising production costs between those which are "variable" (costs that change when the production output changes) and those that are "fixed" (costs not directly related to the volume of production).
Total variable and fixed costs are compared with sales revenue in order to determine the level of sales volume, sales value or production at which the business makes neither a profit nor a loss (the "break-even point").

THE BREAK-EVEN CHART

In its simplest form, the break-even chart is a graphical representation of costs at various levels of activity shown on the same chart as the variation of income (or sales, revenue) with the same variation in activity. The point at which neither profit nor loss is made is known as the "break-even point" and is represented on the chart below by the intersection of the two lines:

Breakeven analysis is useful for the following reasons:

- It helps to determine remaining/unused capacity of the concern once the breakeven is reached. This will help to show the maximum profit on a particular product/service that can be generated.
- It helps to determine the impact on profit on changing to automation from manual (a fixed cost replaces a variable cost).
- It helps to determine the change in profits if the price of a product is altered.
- It helps to determine the amount of losses that could be sustained if there is a sales downturn.

Ways to monitor Breakeven point

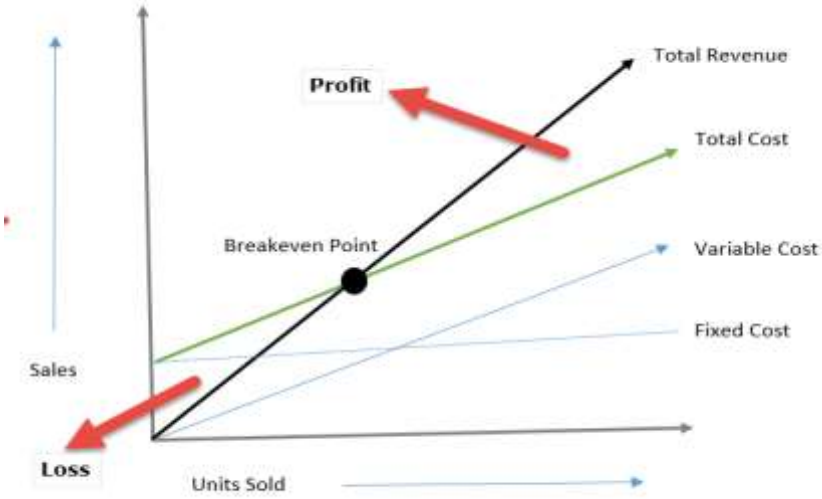
- **Pricing analysis:** Minimize or eliminate the use of coupons or other price reductions offers, since such promotional strategies increase the breakeven point.
- **Technology analysis**: Implementing any technology that can enhance the business efficiency, thus increasing capacity with no extra cost.
- **Cost analysis:** Reviewing all fixed costs constantly to verify if any can be eliminated can surely help. Also, review the total variable costs to see if they can be eliminated. This analysis will increase the margin and reduce the breakeven point.
- **Margin analysis:** Push sales of the highest-margin (high contribution earning) items and pay close attention to product margins, thus reducing the breakeven point.
- **Outsourcing:** If an activity consists of a fixed cost, try to outsource such activity (whenever possible), which reduces the breakeven point.

Benefits of Break-even analysis

- **Catch missing expenses:** When you're thinking about a new business, it's very much possible that you may forget about few expenses. Therefore, if you do a break-even analysis you have to review all your financial commitments to figure out your break-even point. This analysis certainly restricts the number of surprises down the road.
- **Set revenue targets:** Once the break-even analysis is complete, you will get to know how much you need to sell to be profitable. This will help you and your sales team to set more concrete sales goals.
- **Make smarter decisions:** Entrepreneurs often take decisions in relation to their business based on emotion. Emotion is important i.e. how you feel, though it's not enough. In order to be a successful entrepreneur, your decisions should be based on facts.
- **Fund your business:** This analysis is a key component in any business plan. It's generally a requirement if you want outsiders to fund your business. In order to fund your business, you have to prove that your plan is viable. Furthermore, if the analysis looks good, you will be comfortable enough to take the burden of various ways of financing.
- **Better Pricing:** Finding the break-even point will help in pricing the products better. This tool is highly used for providing the best price of a product that can fetch maximum profit without increasing the existing price.

- **Cover fixed costs:** Doing a break-even analysis helps in covering all fixed cost.

Graphically



BASIC CONCEPTS

- **Variable Costs:** These are costs that change with the production levels or sales. Examples include: Costs of materials used in the production of goods.
- **Fixed Costs:** These are costs that remain the same regardless of sale/Output levels. Examples include: Rent, Insurance and wages.
- **Revenue:** Total income received
- **Profit:** The money you have after subtracting fixed and variable cost from revenue.
- **Volume** is the level of production, normally expressed as the number of units produced and sold.
- **Cost** is categorized into *fixed costs* and *variable costs.* Fixed costs remain constant regardless of the number of units produced. Variable costs vary with the volume of units produced e.g. labor, materials.

Mathematically they can be expressed as:
Total cost = Fixed Cost + Total Variable Cost
TC = FC + TVC, Where TVC= ***VC*Q***
Total revenue = Volume (Quantity) * price
Profit = Total revenue – Total cost

Where:
FC = Fixed cost
Q = Volume (i.e. number of units produced and sold)
VC = variable costs per unit
P = Price per unit
There is need to determine how much we need to produce to earn a profit. The point where total revenue is equal to total cost is the break-even point. At any volume above the break-even point, a profit is earned.

Break-even point is equal;

TR = TC

TC = FC+ VC*Q

Therefore

TR = (FC + VC*Q)

P*Q = (FC + VC*Q)

Q (P – VC) = FC

<u>Q = FC/ (P-VC)</u>

Example One:

Shumuk ltd is planning to begin producing rubber tyres for its motorcycles instead of outsourcing them. The initial investment in plant and equipment needed for production of rubber tyres is estimated to be UGX 20 million. Labor and materials costs are approximated at UGX 50,000 per tyre. Assuming that each tyre can be sold at a price of UGX 100,000, determine the volume of demand necessary to break-even.

Solution:

Fixed cost = UGX 20milliom

Variable cost = UGX 50,000 per tyre

Expected Price per tyre = UGX 100,000

To calculate break-even point

Q = FC/(P-VC) = 20,000,000/ (100,000 – 50,000) = 20,000,000/50,000= 400 tyres

Using break-even analysis to select a process among alternatives:

Example Two:

Shumuk ltd is contemplating to make a larger initial investment of UGX 100 million for more automated equipment that will reduce the variable cost of manufacture to UGX 20.000 per tyre. Comparing the old process with the new process, the demand of volume to be chosen for each process would be:

Solution: Equate total costs of process A and Process B

20,000,000+ 50,000**Q** = 100,000,000 + 20,000**Q**

30,000**Q** = 80,000,000

<u>Q = 2,667 tyres</u>

If demand is less than **2,667 tyres**, the alternative with the lowest fixed cost ***(process A)*** should be chosen. If demand is greater than or equal to **2,666 tyres,** the lowest variable cost ***(process B)*** should be chosen.

Practice questions

(a) Explain any five (5) dimensions that can be used in measuring service quality.

(b) As consumers, we buy millions of products every year, Just like human beings, these products undergo a life cycle. As a business student, discuss the different life cycle stages of any product of your choice.

(c) Describe the role of production and operations management in providing value to customers in both manufacturing and service organisations.

(d) For over two century's Operations and Production Management has been recognized as an important factor in a country's economic growth. Briefly explain the evolution historical developments of Operations Development to date.

(e) In one of the Top Management meetings of Emma manufacturing company, the head of operations suggested that there's a need to re-design the current layout of the organization. Other members of the team where not convinced and asked him to write a report about this sating possible reasons for the suggestion for a discussion in the next meeting.

Required: Describe the possible reasons that the head of operations would include in the report.

(f) Distinguish between process and fixed position layout clearly showing their advantages and limitations.